Sustainable Aviation Fuel Supply Roadmap for Korea

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Objective and contribution

SAF supply planning requires matching domestic feedstocks and conversion routes with future fuel demand and the carbon and hydrogen needed for synthesis. I led an eight-member course team in developing a Korean supply roadmap, using European blending targets as a scenario benchmark.

Supply and demand analysis

We compared hydroprocessed esters and fatty acids (HEFA), alcohol-to-jet (ATJ), gasification/Fischer–Tropsch (FT), and power-to-liquid (PtL) pathways. To estimate supply potential from domestic resource inventories, we used feedstock heating values and conversion efficiencies for bio-SAF and carbon-content calculations for synthetic SAF. We projected jet-fuel demand through 2050 using linear regression on 2009–2021 consumption data and applied the benchmark blending targets.

Table 1. Estimated annual SAF potential from domestic resources.

Resource or carbon source	Route	Annual potential (million L/year)
Rice straw	ATJ	506.8 (bio-SAF)
Used cooking oil	HEFA	227.8 (bio-SAF)
Organic wastes and sewage sludge	PtL	268.0 (synthetic SAF)

2050 synthetic-SAF balance (million L/year): supply 268.0; requirement 522.4; shortfall 254.3. Values are rounded independently from the report's unrounded calculations.

Under the report's assumptions, the combined bio-SAF potential of 734.6 million L/year covered the modeled requirement through 2050. The synthetic-SAF potential met the modeled requirement through 2045 but left a projected shortfall of 254.3 million L/year in 2050, prompting consideration of additional carbon resources.

Supply conditions and project outcome

We examined direct air capture (DAC) as an additional CO_2 source and considered hydrogen availability. We compared projected SAF prices with a conventional jet-fuel price benchmark to examine the cost gap over time. The roadmap highlighted feedstock collection systems, additional CO_2 sourcing, and hydrogen supply as conditions for domestic production. The project ranked first among eight teams.